

ARK-460 — Results

Cloud IAM Role Grant · Dependency Loss

Experiment ID: ARK-460 · **Institution:** Remnant Fieldworks Inc. · **PI:** Derek Adam Hone
Substrate: Classical software · **Guards:** Dual independent (V1 JS, V2 Python)
Official execution: 2026-07-18 (post-lock)

Verdict:  **PASS**

When critical dependencies are **UNAVAILABLE** at execution time, the gate correctly **fails safe** (HOLD — cannot confirm validity) rather than wrongly ALLOWing.

Criterion	Requirement	Value	Result
C1 g_allow_available	≥ 0.95	1.0000	 PASS
C2 g_hold_unavailable_min	≥ 0.95	1.0000	 PASS
C3 g_margin	≥ 0.90	0.9500	 PASS

- Dual-guard concordance: **800/800 = 100.00%**
- Kill-gate: wrongly ALLOWs **150/150 = 100%** of HOLD arms (testbed falsifiable ✓)

Per-Arm Results

Arm	Label	Expect	Correct	Concordance
1	ALL-AVAILABLE	ALLOW	100/100	100/100
2	PolicyRegistry UNAVAILABLE	HOLD	100/100	100/100
3	IdentityProvider UNAVAILABLE	HOLD	100/100	100/100
4	SignatureVerifier UNAVAILABLE	HOLD	100/100	100/100
5	AuditLogger UN- AVAILABLE	HOLD	100/100	100/100
6	MULTIPLE UN- AVAILABLE	HOLD	100/100	100/100
7	ALL-UNAVAIL- ABLE	HOLD	100/100	100/100
8	ALL-AVAILABLE- baseline	ALLOW	100/100	100/100

Total: 800 decisions, all 100% correct, 100% dual-guard concordance.

Findings

1. **Perfect fail-safe behavior.** Arms 2–7 (at least one dependency unavailable) all HOLD 100/100. No ALLOW when dependencies are down.
2. **ALLOW only when all dependencies available.** Arms 1, 8 both ALLOW 100/100 when all dependencies available.
3. **Independent dual guards in perfect agreement.** V1 (JS) and V2 (Python) produced identical decisions on all 800 scenarios.

Integrity

- Locked files unchanged (verified via `./compute_hashes.sh`).
- Both guards ran clean on first execution; no post-lock harness fix required.

Honest bounds

Classical software test of authorization control logic. This tests the **dependency-loss logic**; ARK-458 tested exact-action-binding, ARK-459 tested revocation. **No claim** that this legally validates any patent claim or certifies RF-100 conformance.

Published under the Remnant Fieldworks Standing Covenant. To God be the glory. Proof Before Power.
Verification Before Execution.